

SAT 7

Math Module 1

Question #

ID

SAT7 M 1.1

a07ma101

Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid \$2 for the box of cereal and \$1.90 per pound for the strawberries. If Lorenzo paid a total of \$9.60 for the box of cereal and the strawberries, which of the following equations can be used to find p , the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

(A) $1.90p + 2 = 9.60$

(B) $1.90p - 2 = 9.60$

(C) $1.90 + 2p = 9.60$

(D) $1.90 - 2p = 9.60$

SAT7 M 1.2

a07ma102

The function f is defined by $f(x) = 25x + 30$. What is the value of $f(x)$ when $x = 2$?

(A) 50

(B) 57

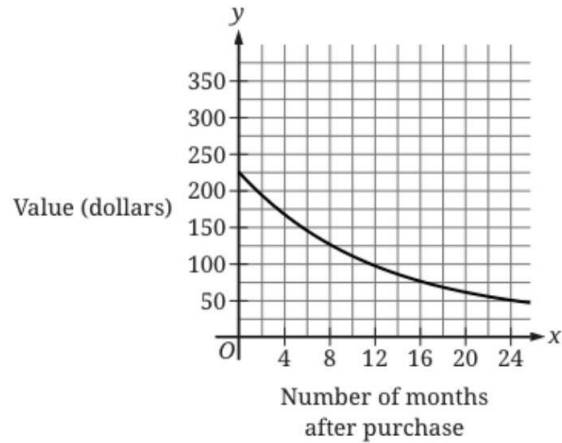
(C) 80

(D) 110

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Question # ID
SAT7 M 1.3 a07ma103



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

- (A) The estimated value of the tablet was \$225 when it was purchased.
- (B) The estimated value of the tablet 24 months after it was purchased was \$225.
- (C) The estimated value of the tablet had decreased by \$225 in the 24 months after it was purchased.
- (D) The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

SAT7 M 1.4 a07ma104

$$|p| + 61 = 65$$

Which value is a solution to the given equation?

- (A) $\frac{65}{61}$
- (B) 4
- (C) 126
- (D) 130

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SAT7 M 1.5 a07ma105

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

(A) 20°

(B) 45°

(C) 135°

(D) 160°

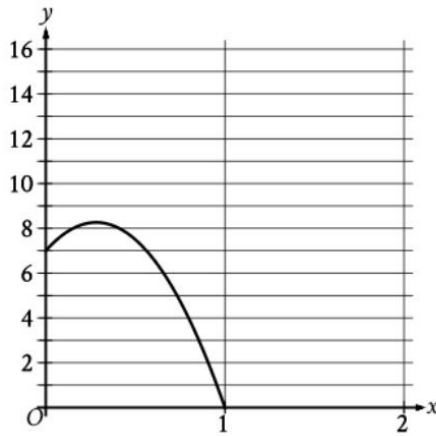
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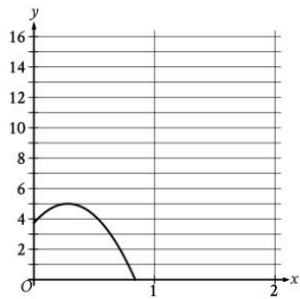
SAT7 M 1.6 a07ma106

During the first part of an experiment, a ball was launched from a 7-foot-tall platform. The graph shows the height y , in feet, of the ball x seconds after it was launched during the first part of the experiment.

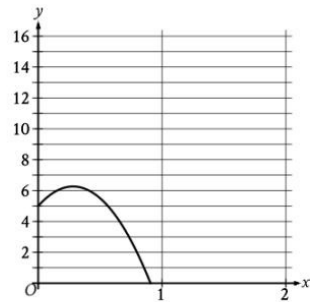


During the second part of the experiment, the ball was launched the same way, but from a platform that is 2 feet shorter than the first platform. Which of the following graphs could represent the height y , in feet, of the ball x seconds after it was launched during the second part of the experiment?

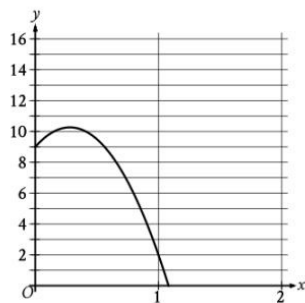
(A)



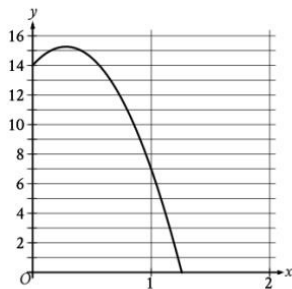
(B)



(C)



(D)



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SAT7 M 1.7 a07ma107

The ratio x to y is equivalent to the ratio 9 to 5. If the value of x is 162, what is the value of y ?

SAT7 M 1.8 a07ma108

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won S square tokens and C circle tokens worth a total of 1,120 points. The equation $80S + 90C = 1,120$ represents this situation. How many more points is a circle token worth than a square token?

(A) 950

(B) 90

(C) 80

(D) 10

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Question # ID
SAT7 M 1.9 a07ma109

$$\frac{(x+9)(x-9)}{x+9} = 7$$

What is the solution to the given equation?

(A) 7

(B) 9

(C) 16

(D) 63

SAT7 M 1.10 a07ma110

The equation $12t + b = c$ relates the variables t , b , and c . Which of the following correctly expresses the value of $c - b$ in terms of t ?

(A) $\frac{t}{12}$

(B) t

(C) $t + \frac{1}{12}$

(D) $12t$

SAT7 M 1.11 a07ma111

What is an x -coordinate of an x -intercept of the graph of $y = 3(x - 14)(x + 5)(x + 4)$ in the xy -plane?

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Question # ID
SAT7 M 1.12 a07ma112

The function f is defined by $f(x) = 5\left(\frac{1}{4} - x\right)^2 + \frac{11}{4}$. What is the value of $f\left(\frac{1}{4}\right)$?

SAT7 M 1.13 a07ma113

A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

(A) The center is at $(13, k)$ and the radius is 8.

(B) The center is at $(k, 13)$ and the radius is 8.

(C) The center is at $(k, 13)$ and the radius is 64.

(D) The center is at $(13, k)$ and the radius is 64.

SAT7 M 1.14 a07ma114

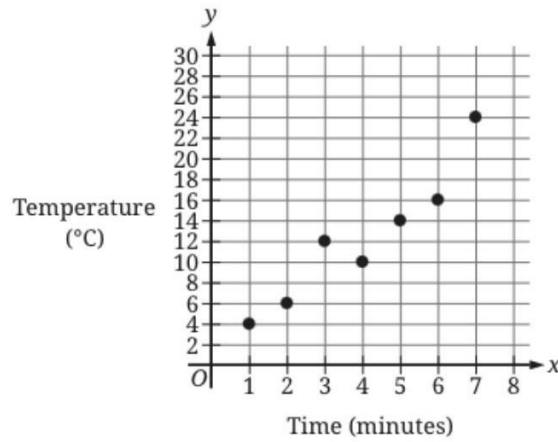
A circle has a radius of 2.1 inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

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Question # ID
SAT7 M 1.15 a07ma115

During a study, the temperature, in degrees Celsius ($^{\circ}\text{C}$), of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature y , in $^{\circ}\text{C}$, of the air in the chamber x minutes after the start of the study.



What was the average rate of change, in $^{\circ}\text{C}$ per minute, of the recorded temperature of the air in the chamber from $x = 5$ to $x = 7$?

SAT7 M 1.16 a07ma116

$$\begin{aligned} 3y &= 4x + 17 \\ -3y &= 9x - 23 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $39x$?

(A) -18

(B) -6

(C) 6

(D) 18

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SAT7 M 1.17

a07ma117

The function $f(t) = 40,000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

☐ (A) 2☐ (B) 790☐ (C) 1,580☐ (D) 40,000

SAT7 M 1.18

a07ma118

If $5 - 7(2 - 4x) = 16 - 8(2 - 4x)$, what is the value of $2 - 4x$?

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Question # ID
SAT7 M 1.19 a07ma119

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for 46 days. On February 15, there were 99 dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January 1 to February 15 was 12.50%. How many dragonflies were in the habitat on January 1?

(A) 88

(B) 87

(C) 12

(D) 8

SAT7 M 1.20 a07ma120

$$2x^2 - 8x - 7 = 0$$

One solution to the given equation can be written as $\frac{8-\sqrt{k}}{4}$, where k is a constant. What is the value of k ?

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SAT7 M 1.21

a07ma121

The function $f(x)$ is defined as 19 more than 4 times a number x . If $y = f(x)$ is graphed in the xy -plane, what is the best interpretation of the x -intercept?

- (A) When $f(x) = 0$, the number is $-\frac{19}{4}$.
- (B) When the number is 0, $f(x) = 19$.
- (C) The value of $f(x)$ increases by 1 for each increase of 4 in the value of the number.
- (D) For each increase of 1 in the value of the number, $f(x)$ increases by 4.

SAT7 M 1.22

a07ma122

Which of the following expressions is equivalent to $(\sin 24^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\sin 66^\circ)$?

- (A) $2(\cos 66^\circ)(\sin 24^\circ)$
- (B) $2(\cos 66^\circ) + 2(\cos 24^\circ)$
- (C) $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$
- (D) $(\cos 66^\circ)^2 + (\sin 24^\circ)^2$